



Head-to-head comparison of four plasma neurofilament light chain (NfL) immunoassays

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ABSTRACT

Background: Neurofilament Light Chain (NfL) is an emerging blood biomarker of neuro-axonal injury and neurodegeneration with the potential to be used in the clinical management of various neurological conditions. Various NfL immunoassays are in development on high-throughput automated systems, but little information is available related to the comparability between assays. In this study, we performed a head-to-head comparison of four NfL immunoassays using plasma samples from individuals with various neurological conditions.

Methods: EDTA plasma samples in which NfL was ordered clinically were stratified according to diagnosis. NfL concentrations (pg/mL) in plasma were obtained using the Quanterix Simoa®, the Roche Elecsys, the Siemens Healthineers Atellica®IM, and the Fujirebio Lumipulse® NfL assays. Passing-Bablok regression analyses were performed to assess the correlation and bias between methods. Additionally, the distribution of NfL concentrations for each assay was assessed in three disease groups: amyotrophic lateral sclerosis (ALS) upon initial diagnosis, ALS treated, and multiple sclerosis (MS).

Results: The R^2 between assays were all ≥ 0.95 , however, significant proportional bias was observed between some assays. In particular, the Roche Elecsys assay NfL concentrations were significantly lower ($\sim 85\%$) when compared against the other three assays. The four assays were comparable with regards to the percentage of patients that were identified as having an elevated NfL result in the various clinical groups: ALS initial diagnoses (83–94%), ALS untreated (93–100%), and MS (8–18%).

Conclusions: This is the first study describing a head-to-head comparison of four automated NfL immunoassays. We demonstrate that there is a strong correlation between assays but a lack of standardization which is evident by the bias observed between some of the evaluated methods. These analytical differences will be important to consider when using NfL as a biomarker of neurodegeneration.

1. Introduction

Neurofilament light chain (NfL) is a neuron-specific cytoskeletal protein that has been shown to be a useful biomarker of neuro-axonal injury. Concentrations of NfL in cerebrospinal fluid (CSF) and blood increase proportionally to the degree of axonal damage in a variety of neurological disorders including inflammatory, neurodegenerative, traumatic, and cerebrovascular diseases [1]. Consequently, NfL has been studied as a biomarker of disease activity and/or prognosis in multiple sclerosis (MS), Alzheimer's disease (AD), frontotemporal dementia (FTD), amyotrophic lateral sclerosis (ALS), atypical parkinsonian disorders (APD) and traumatic brain injury (TBI) among others [2,3]. As

evidence continues to grow, there has been increased interest in using NfL in the clinical and research setting as a diagnostic, prognostic, and monitoring biomarker for neurological diseases. Over 100 drug trials for neurodegenerative diseases have included NfL measurements in blood or CSF as primary or secondary endpoints of drug efficacy [4]; and most recently the Food and Drug Administration (FDA) granted accelerated approval of tofersen, a drug for *SOD1*-associated ALS, based on its NfL-lowering effect in blood [5].

Circulating NfL concentrations in blood have been reported to increase with age. We have previously reported an overall yearly increase of approximately 2–3% in healthy controls [6]. Another study reported a nonlinear increase with a yearly median increase of 0.9% in 40–50

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year-olds, and 2.7 % in 50 – 60 year-olds. After 60 years of age, the yearly median increase in serum NfL reached a maximum slope of up to 4.3 %, whereas after age 70 the slope remained stable [3]. Benkert et al showed that NfL increased exponentially with age and at a steeper increased age after 50 years of age [7]. The results of the studies mentioned above highlight the importance of having well-characterized reference intervals that account for this age-related increase for the useful clinical interpretation of NfL during the evaluation of neurological disorders. A limited number of studies using well-characterized adult cohorts have reported age-specific reference intervals for plasma NfL using the Quanterix Simoa assay [6,8,9]. However, reference intervals for newer assays are not available in the literature or from the assay's manufacturers.

Various immunoassays for the quantitation of NfL are currently in development and clinical laboratories are validating research-use-only (RUO) kits and offering these assays as laboratory-developed tests (LDTs). However, the lack of a certified reference material and reference method for NfL prevents the comparison of results between immunoassays from different manufacturers [10]. Furthermore, there are limited published data on the expected differences or biases between immunoassay methods. The objective of this study was to evaluate and compare the performance of the following assays in plasma: Quanterix Simoa® NF-Light Advantage kit, Roche Elecsys NFL hs, Siemens Healthineers Atellica® IM Neurofilament Light Chain (NfL), and Fujirebio Lumipulse® G NfL Blood.

2. Material and methods

2.1. Samples

EDTA plasma samples (N = 122) from adult patients (>18 years) who had given research authorization, were seen at Mayo Clinic from 04/28/22 to 06/18/22, had a plasma NfL ordered clinically, and had testing performed at Mayo Clinic laboratories were included in this study. Samples, which had undergone one previous freeze–thaw cycle, were aliquoted and stored frozen at –20 °C until testing. Samples that exhibited a > 150 % difference between methods were repeated and evaluated for interferences. Based on this, eight samples were excluded from the final analysis due to suspicion of heterophilic antibody interference after sample testing using heterophile blocking tubes (Scantibodies cat#3IX762) led to significantly different results on one or more of the evaluated assays. For the remaining 114 unique patients, the

electronic medical records were reviewed, and samples were classified into the following clinical disease groups: MS (n = 39), ALS (initial diagnosis [n = 18], treated ALS [n = 14], and probable ALS [n = 2]), neuropathies (n = 11), primary lateral sclerosis (n = 5), and dementia (n = 2). The patients who did not fit into one of these clinical categories were classified as “other” (n = 23). This group included a mixture of diagnoses such as slow motor neuron disease, Erdheim-Chester Disease, fasciculation, myelopathy, neuromyelitis optica, neurosarcoidosis, normal pressure hydrocephalus, paraparesis, radiculopathy, and tumefactive demyelinating lesion. The MS group was very heterogeneous with regards to treatment or MS classification and thus was not further categorized into specific groups. The cohort (N = 114) was comprised of 62 (54 %) females and 52 (46 %) males. The median [IQR] age was 57 [47–67] years. This study was reviewed and approved by the Mayo Clinic Internal Review Board. All work has been carried out in accordance with The Code of Ethics of the World Medical Association (Declaration of Helsinki) for experiments involving humans.

2.2. Immunoassays

Descriptions of the assays included in this study are shown in Table 1. NfL (pg/mL) was measured per manufacturer instructions using the Quanterix Simoa HD-X analyzer (Simoa NF-Light V1 Advantage kit), the Roche Cobas 8000 (Elecsys NFL hs), and the Fujirebio LUMIPULSE G System (Lumipulse® G NfL Blood) at Mayo Clinic, Rochester, MN. The NfL assay (Simoa NF-Light Advantage) performed at Mayo Clinic Laboratories is considered a laboratory-developed test and its performance characteristics were determined by Mayo Clinic in a manner consistent with CLIA requirements. Based on this internal analytical validation, the Simoa mean intra- (N = 20) and inter-assay (N = 20 over 10 days) %CV (range) were as follows: 4.6 % (3.4–6.0%) concentration range (CR) 3.9–240 pg/mL, and 8.5 % (6.0–12.0%); CR 3.6–235 pg/mL, respectively. The Elecsys and Lumipulse assays are research-use-only (RUO) assays. For the Elecsys assay, in-house studies showed a mean intra-assay (N = 20) %CV (range) of 2.2% (1.0–3.6%); CR 3.2–348 pg/mL and a mean interassay (N = 10, 5 days) of 4.2 % (2.6–6.2%); CR 3.3–358 pg/mL. For the Lumipulse assay, manufacturer quality control (QC) material, ran over 3 days, showed mean concentrations of 21.6 pg/mL (2.3% CV) and 768.5 pg/mL (1.3% CV). Sample aliquots were sent to Siemens Healthcare Diagnostics, Inc. (Tarrytown, New York) for measurement of NfL using the Siemens Healthineers Atellica IM NfL RUO assay. QC material, ran over 6 days, showed a mean %CV (range) of

Table 1
NfL assay characteristics based on published information.

Manufacturer	Quanterix Corporation	Roche Diagnostics	Siemens Healthcare Diagnostics	Fujirebio Diagnostics
Instrument	Simoa® HD-X Analyzer™	Cobas® 8000 e 801	Atellica® IM Analyzer	Lumipulse® G1200
Assay Name	Simoa® NF-Light Advantage kit	Elecsys NFL hs	Atellica® IM Neurofilament Light Chain (NfL)*	Lumipulse® G NfL Blood
LOD	0.085 pg/mL	0.140 pg/mL	1.3 pg/mL	Not available
Precision studies %CV	Internal Validation: Intra-assay: <or = 4.63 % Inter-assay: <or = 8.46 %	Collier et al. [18] Intra-assay: <or = 5.2 % CV, Inter-assay: <or = 7.2 %	Merabet [24] Intra-assay: <or = 6.4 % Inter-assay: <or = 12.4 %	Not available
Analytical Measuring Range (per manufacturer)	0.345–360 pg/mL reportable range (x4 dilution) 0.345–1440 pg/mL	0.168–1000 pg/mL	2.5–300 pg/mL reportable range (x10 dilution) 2.5–3000 pg/mL	2.0–5000 pg/mL
Antibody pair	Uman clone 2.1 and 47.3 mouse monoclonal antibodies	Proprietary	Uman clone 2.1 and 47.3 mouse monoclonal antibodies	Mouse monoclonal antibodies
Sample types listed in package insert	Serum, EDTA Plasma, CSF	Serum, Plasma, CSF	Serum, EDTA Plasma	Serum, K2EDTA Plasma, Li Hep Plasma

*In development, not available for sale.

4.5% (4.2–5.0%); CR 9.9–148 pg/mL. For all assays, testing personnel were blinded to clinical diagnosis and/or clinically reported NFL results.

The classification of NFL concentrations as elevated or not elevated was based on a designated reference interval for each assay. See Supplementary Table 2 for detailed reference interval information. For Simoa NFL, age-specific reference intervals published from an in-house study were used [6]. Age-specific NFL reference intervals for the Elecsys NFL assay were obtained from LabCorp (<https://www.labcorp.com/tests/140455/neurofilament-light-chain-serum>; accessed November 30, 2023) [11]. Atellica age-specific NFL 97.5th reference intervals were based on the NHANES model [12] and were determined using the online calculator (https://brbdai-kathryn-fitzgerald.shinyapps.io/sNFL_dashboa rd/ accessed November 30, 2023) [13]. Lumipulse NFL reference intervals have not been published. Therefore, we derived Lumipulse NFL reference intervals based on the Passing-Bablok regression equation between Simoa and Fujirebio (n = 114). Of note, published reference intervals for the Elecsys and Atellica NFL assays were based on serum samples. However, because only small differences are expected between serum and plasma, the usage of serum-derived reference intervals should have minimal impact on this comparison [14].

2.3. Statistical analyses

Statistical analysis was performed using Analyse-it® version 6.15.4 for Microsoft Excel. Passing-Bablok regression analysis and the coefficient of determination (R^2) were performed for all combinations of assays. The percentage of samples with an elevated NFL for each assay, within three disease groups (ALS initial diagnosis, ALS treated, and MS)

were calculated as described above.

3. Results

Passing-Bablok regression analyses for all NFL assay combinations are shown in Fig. 1. When using the Simoa NFL assay as the reference (x-axis) (Fig. 1a), the assays showed excellent correlation ($R^2 \geq 0.95$), however, significant biases were observed. The Elecsys NFL assay showed the largest bias with results being 84 % lower than the Simoa assay results (slope of 0.16, 95 % CI: 0.15–0.17). The Atellica assay results were 13 % lower than the Simoa assay results (slope of 0.87, 95 % CI: 0.83–0.91); whereas the Lumipulse assay showed the least bias from all three assays with results matching the Simoa assay within 6 % (slope of 1.06, 95 % CI: 1.01–1.09).

Using a subset of samples that included individuals with ALS (initial diagnosis or treated) and MS, the distribution of NFL concentrations amongst the assays was evaluated (Fig. 2). In all 4 assays, NFL median concentrations (pg/mL) were higher in the ALS initial diagnosis patients with no treatment compared to the ALS treated group (Simoa 94.5 (95 % CI: 65.8–121.0) vs 56.8 (95 % CI: 33.6–105.0); Elecsys 14.7 (95 % CI: 11.2–23.1) vs 9.5 (95 % CI: 6.7–19.5); Atellica 87.4 (95 % CI: 55.7–114.1) vs 57.0 (95 % CI: 38.4–100.0); Lumipulse 99.9 (95 % CI: 62.7–140.2) vs 61.7 (95 % CI: 39.1–131.2). Samples from MS patients showed the lowest NFL median concentrations (pg/mL) (Simoa 12.2 (95 % CI: 8.8–15.6); Elecsys 1.95 (95 % CI: 1.7–2.4), Atellica 13.3 (95 % CI: 10.2–15.3); Lumipulse 14.2 (95 % CI: 11.0–16.1). Notably, the MS group represents a very heterogeneous cohort of treated and untreated patients which included the following clinical classifications: MS early

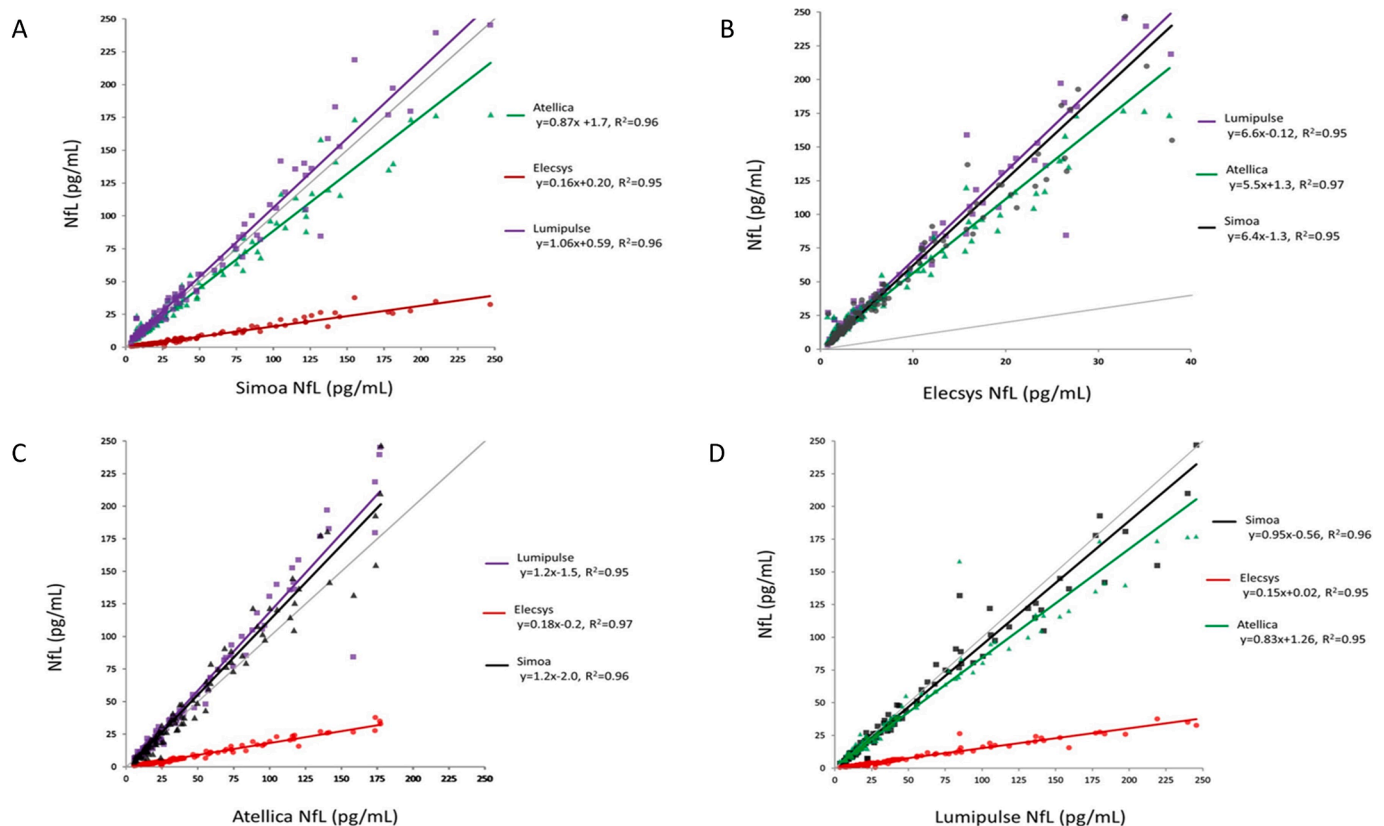


Fig. 1. NFL method comparisons using Passing-Bablok regression analysis. a) Simoa NFL (x-axis) and Lumipulse, Elecsys, and Atellica NFL concentrations (y-axis). Data points and regression lines are color-coded: red circles (Elecsys), green triangles (Atellica), and purple squares (Lumipulse). b) NFL (pg/mL) (x-axis) and Lumipulse, Atellica, and Simoa NFL concentrations (y-axis). Data points and regression lines are color-coded: green triangles (Atellica), purple squares (Lumipulse), and black circles (Simoa). c) Atellica NFL (x-axis) Lumipulse, Simoa, and Elecsys NFL concentrations (y-axis). Data points and regression lines are color-coded: purple squares (Lumipulse), red circles (Elecsys), and black triangles (Simoa). d) Lumipulse NFL (x-axis) Simoa, Elecsys, and Atellica NFL concentrations (y-axis). Data points and regression lines are color-coded: black squares (Simoa), red circles (Elecsys), and green triangles (Atellica). The gray line represents the line of identity. The 95% confidence interval for all slopes are shown in Supplementary Table 1.

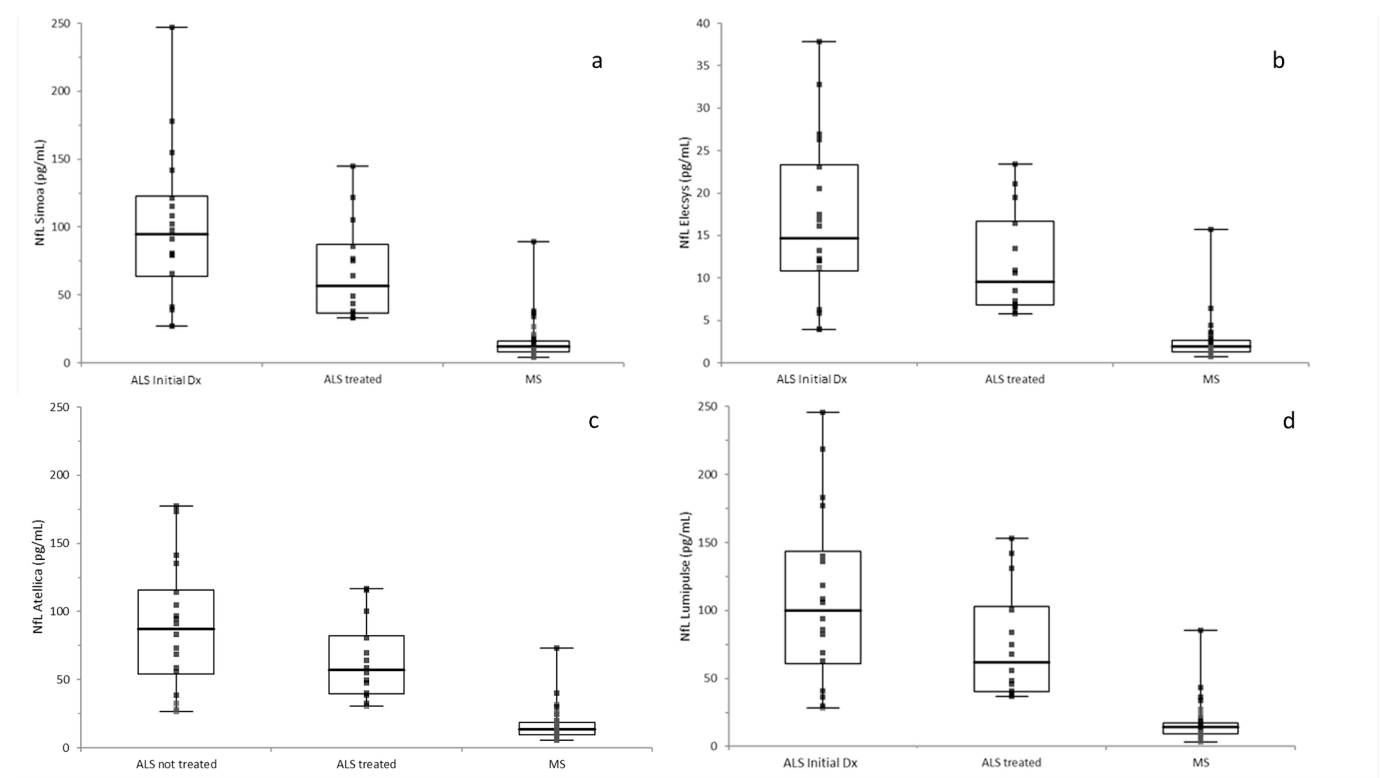


Fig. 2. NfL concentrations in various disease groups. NfL concentrations (pg/mL) for samples from individuals with ALS initial diagnosis (n = 18), ALS treated (n = 14), and MS (n = 39) for Simoa (a), Elecsys (b), Atellica (c) and Lumipulse (d).

state, stable MS, primary progressive MS without disease activity, secondary progressive MS with or without disease activity, and relapsing-remitting MS with and without disease activity. Eighteen of the 39 MS patients (46 %) were undergoing treatment. Using the reference intervals for each assay, NfL results were classified as elevated if the result was above the age-specific reference interval. Table 2 shows the percentage % (exceeding the upper reference limit) for each disease group across all four assays. The four assays were comparable with regards to the percentage elevated for samples categorized as ALS initial diagnoses (83–94 %), ALS untreated (93–100 %), and MS (8–18 %). For the Elecsys and Atellica assays, the number of samples with elevated results differed in some of the groups when using derived reference intervals based on the method comparison equations of the assays using Simoa as the reference (see Supplementary Tables 3 and 4).

4. Discussion

A growing number of NfL immunoassays are in development and available for research use, however, limited information is available regarding the comparability of results between assays. This study showed that while there is a high correlation ($R^2 > 0.95$) between four different NfL assays (Simoa, Elecsys, Lumipulse, and Atellica), there is a

proportional bias between the methods, with regression slopes ranging from 0.16 to 1.05 when comparing against the Simoa assay (x-axis). To our knowledge, this is the first report of a head-to-head comparison of these four immunoassays.

The bias observed between assays is not unexpected as there is no certified reference material for NfL, hence the assays are not standardized. In addition to differences in assay calibrators, antibodies, and detection methods may also contribute to the observed biases. The Simoa and Atellica assays use the same antibody pair (Uman antibodies) but not the same calibrator material, which likely explains the small proportional bias observed between these two assays. While the Lumipulse assay does not use the Uman antibodies (per manufacturer communication), this assay showed the least bias suggesting that the manufacturer likely anchored their calibration to align with the Simoa assay. Unfortunately, no information was available regarding the antibody pair used in the Elecsys assay. The Uman antibodies' epitopes are presumed to be epitopes that are only unmasked upon neurodegeneration-induced proteolysis of the neurofilament. Thus, the Uman-type antibodies could not detect healthy intact neurofilaments [15]. The tight correlation between all assays suggests that antibodies with similar epitopes as the Uman antibodies might be used in all the assay designs.

While limited data are available comparing these platforms, the findings from our study are consistent with NfL data from the Alzheimer's Association Quality Control (QC) program (<https://www.gu.se/en/neuroscience-physiology/the-alzheimers-association-qc-program-for-csf-and-blood-biomarkers>, website accessed Dec 11, 2023) [16]. For example, in a total of 29 participants reporting plasma NfL results (round 42), 4 participants used the Lumipulse assay, and 25 participants used the Simoa assay. The fold differences (Lumipulse/Simoa) in the two samples provided per challenge were 1.26 and 1.06. The fold difference based on samples from our cohort of 114 patients was 1.07. Method comparison studies between Atellica and Simoa serum NfL have been previously reported to have a strong correlation, with an

Table 2
Percentage of samples classified as having an elevated NfL for each of three disease groups.

	ALS Initial Dx (N = 18)	% elevated	ALS treated (N = 14)	% elevated	MS (N = 39)	% elevated
Simoa	17	94 %	14	100 %	7	18 %
Elecsys	16	89 %	14	100 %	7	18 %
Atellica	15	83 %	13	93 %	3	8 %
Lumipulse	17	94 %	14	100 %	7	18 %

intra-class correlation coefficient of 0.87 [17] and a Spearman's rank correlation coefficient of 0.98 [10]. Reported comparisons between Elecsys NfL and the Simoa NfL assays also showed the high correlation and large bias observed in our study [18,19].

The observed bias between NfL assays will preclude the comparability of results between some assays for clinical purposes and the potential use of NfL Z-score calculators developed using specific assays [7,12,20,21]. However, the interpretation of assay-specific data can be facilitated by the use of assay-specific reference intervals. Given that NfL concentrations increase by approximately 2–3 % per year in healthy controls, large sample cohorts are required for robust reference interval determination. Alternatively, transference of NfL reference intervals between platforms could be considered for those assays with an $R^2 > 0.98$ [22,23]. To date, most publications that have reported reference intervals have used the Simoa platform [6,8].

For the Atellica assay, there were differences between the NHANES calculator 97.5th upper limit and the Atellica Derived RI (Supplementary Tables 2 and 3), especially for the younger age groups. While the exact reason for these differences is unknown, possible explanations include differences in study participants, study design, and the type of modeling used to derive the intervals. Of note, in the NHANES study, 57.3 % of participants had at least 1 non-neurologic comorbidity (asthma, celiac disease, gout, rheumatoid arthritis, heart failure, coronary artery disease, angina, liver disease, kidney failure, thyroid disorders, hyperlipidemia, diabetes, COPD, diabetes, or cancer) [12], whereas the Mayo Clinic reference interval study excluded the following conditions: known chronic kidney disease (participants had eGFR > 90 mL/min/1.73 m²), neurological conditions, stroke, myocardial infarction, or atrial fibrillation, and a BMI > 30 kg/m² [6].

In this study, when designated assay-specific reference intervals were applied the percentage of elevated results were comparable between the assays. While the clinical information on the cohort used for the method comparison was limited and no claims in clinical performance can be drawn, it would appear all assays will be suitable for clinical use.

One strength of our study was the use of samples in which NfL was ordered as part of the patient's clinical management. This allowed for the evaluation of assay performance in a population representative of samples that will be tested clinically on these assays. The main limitation of this study was the variation introduced by the various reference intervals that were derived from different populations or research studies. Additionally, one of the assays (Lumipulse) did not have an available published reference interval and thus a transformed reference interval was used. Lastly, a limitation of the study is the use of RUO and LDT assays, in which the performance may not be consistent with future FDA-approved versions of the assays. Although these limitations make for a less-than-perfect comparison of the clinical performance of these assays, they also give an idea of the differences to be expected when using these assays and reference limits in a real-world clinical setting.

5. Conclusion

Our study is the first to compare four plasma NfL immunoassays that will likely be used by clinical laboratories in the near future. Having access to these data will allow laboratories to assess comparability between the assays when selecting an assay especially when having to make decisions about the availability of reference intervals for result interpretation. Ongoing efforts towards the development of reference standard materials [10] will potentially allow for better commutability between methods in the future.

Declarations

This study was reviewed and approved by the Mayo Clinic Internal Review Board.

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CRedit authorship contribution statement

S. Ashrafzadeh-Kian: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **D. Figdore:** Writing – review & editing, Data curation. **B. Larson:** . **R. Deters:** . **C. Abou-Diwan:** Writing – review & editing, Data curation. **J. Bornhorst:** Writing – review & editing, Conceptualization. **A. Algeciras-Schminich:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: SL Ashrafzadeh-Kian reports no disclosures. DJ Figdore reports no disclosures. B Larson reports no disclosures. R Deters reports no disclosures. C Abou-Diwan is a salaried employee of Siemens Healthineers. JA Bornhorst has participated on an advisory board for Sunbird Bio and received an honorarium from Roche Diagnostics. A Algeciras-Schminich has participated in advisory boards for Roche Diagnostics, Fujirebio Diagnostics, and Siemens Healthineers.

Data availability

Data will be made available on request.

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Roche Diagnostics provided NfL reagents at no cost to support this testing. Siemens Healthineers performed NfL testing at their facility at no cost.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cca.2024.119817>.

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